



UNIVERSITY
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SOUNDLOG: AN ONLINE MUSIC CATALOG AND PREFERENCE TRACKING WEB APPLICATION

A Web Development 1 Proposal
Presented to the Faculty of the
Department of Computer, Information Sciences and Mathematics
University of San Carlos

In Partial Fulfillment
of the Requirements for
WEB DEVELOPMENT 1

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CHAPTER 1

INTRODUCTION

1.1 Rationale

The digital revolution has fundamentally transformed the manner in which individuals discover, consume, and discuss creative media. Platforms such as Letterboxd have demonstrated the remarkable value of community-driven catalogs for film enthusiasts, enabling users to log their viewing history, assign ratings, and engage in critical discourse. Despite the widespread popularity of music streaming services — most notably Spotify, Apple Music, and YouTube Music — no dedicated web application currently exists that empowers users to maintain a comprehensive, personalized catalog of their music listening experiences in a socially engaged, community-oriented environment. Existing platforms prioritize automated recommendations and algorithmic curation over deliberate, reflective personal logging.

SoundLog is proposed as a response to this identified gap in the digital music ecosystem. As an online music catalog and preference tracking web application, SoundLog is designed to allow users to log albums, tracks, and artists they have listened to; assign star-based ratings; compose personal reviews; and interact with the broader community through likes, comments, and shared lists. Inspired by the model established by Letterboxd in the film domain, SoundLog extends the concept to music, addressing the unmet need for a platform where deliberate music appreciation — rather than passive consumption — is the central organizing principle.

The significance of this system lies not only in its functional utility but also in its potential contribution to music discourse, cultural memory, and personal identity construction through media engagement. By providing a structured yet expressive space

for music logging, SoundLog seeks to serve audiophiles, casual listeners, music critics, and students of music alike.

1.2 Objectives

1.2.1 General Objectives

The general objective of this study is to analyze, design, and document a web-based music catalog system that enables users to track, rate, review, and share their music listening preferences within an interactive social platform.

1.2.3 Specific Objectives

Specifically, this study aims to:

1. Design a user authentication and profile management system that allows users to register, maintain personal profiles, and manage their listening logs securely.
2. Develop a comprehensive music database structure supporting the cataloging of albums, tracks, artists, and genres, with associated metadata.
3. Implement a user-driven rating and review system that permits star-based evaluations and written critiques of music entries.
4. Create a social interaction layer enabling users to follow other users, like reviews, comment on entries, and share curated music lists.
5. Establish search and discovery functionalities that allow users to locate music entries, user profiles, and community reviews efficiently.
6. Ensure the system adheres to established non-functional requirements including performance, security, usability, and scalability standards.

1.3 Beneficiaries

The development and deployment of SoundLog is anticipated to provide measurable benefits to a diverse set of stakeholders. The primary and secondary beneficiaries of the proposed system are identified as follows:

Music Enthusiasts and Audiophiles. The most direct beneficiaries of SoundLog are individuals who actively seek to document and reflect upon their music listening experiences. The platform provides these users with a structured, expressive tool for personal cataloging and self-expression through music.

Casual Listeners. Users who engage with music recreationally but wish to maintain a record of their listening history will benefit from SoundLog's accessible logging interface, which requires no technical expertise.

Music Critics and Reviewers. Individuals engaged in music criticism, whether professionally or as hobbyists, will benefit from a dedicated space to publish reviews and receive community feedback, thereby broadening their audience and critical engagement.

Musicians and Artists. Independent musicians and recording artists may utilize the platform to gauge listener reception of their work, as user-generated reviews and ratings provide unfiltered critical feedback.

Music Educators and Students. Academics and students in the fields of musicology, cultural studies, and media studies may leverage SoundLog as a repository of community-curated opinions and listening trends, serving as a supplementary resource for qualitative music analysis.

Developers and System Analysts. The documentation and analysis presented in this paper serves as a reference for practitioners and researchers interested in community-driven media cataloging systems and social web application design.

1.4 Scope and Limitation

This section delineates the boundaries within which SoundLog is conceived, designed, and evaluated. A clear understanding of the system's scope and limitations is essential to managing expectations and establishing the context for subsequent analytical work.

Scope. SoundLog is scoped as a web-based application accessible via standard browsers on desktop and mobile devices. The system encompasses the following functional domains: user registration and authentication; personal music log management; album, track, and artist entry cataloging; star-based rating and written review submission; social features including following, liking, and commenting; curated list creation and sharing; and search and filtering functionality. The initial design targets music listeners in English-speaking digital environments, with the database structured to accommodate international music entries.

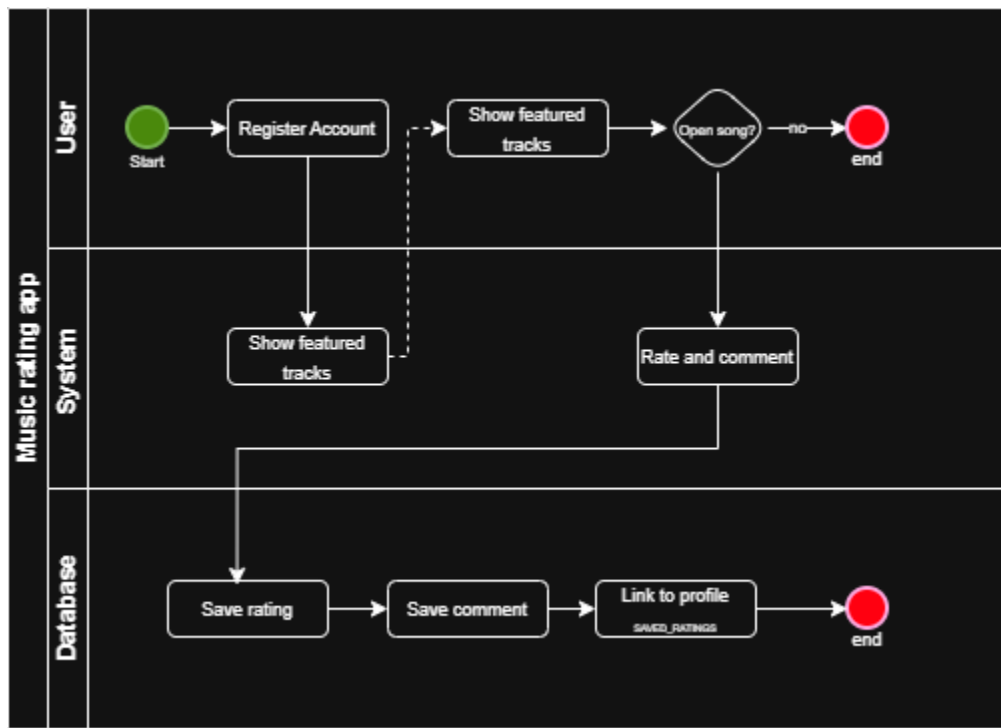
Limitations. The following limitations apply to the current scope of SoundLog's analysis and design. First, the system does not include native music streaming or audio playback capabilities, as it functions exclusively as a cataloging and social review platform. Second, SoundLog does not integrate directly with third-party streaming APIs (e.g., Spotify Web API, Apple MusicKit) in its initial design phase, though such integration is acknowledged as a viable future enhancement. Third, the system's social features are limited to asynchronous interactions; real-time messaging between users falls outside the

present scope. Fourth, the platform is designed for individual user engagement and does not currently accommodate organizational or band accounts. Finally, machine learning-based music recommendation algorithms are considered beyond the scope of this paper and are reserved for future development phases.

CHAPTER 2

ANALYSIS AND DESIGN

2.1 Business Process



2.2 Users

A thorough understanding of the intended user base is fundamental to the design of any information system. SoundLog is conceived to serve a heterogeneous user population, each bringing distinct needs, expectations, and levels of technological proficiency to the platform. The following user categories have been identified through analysis of comparable platforms and the envisioned application context:

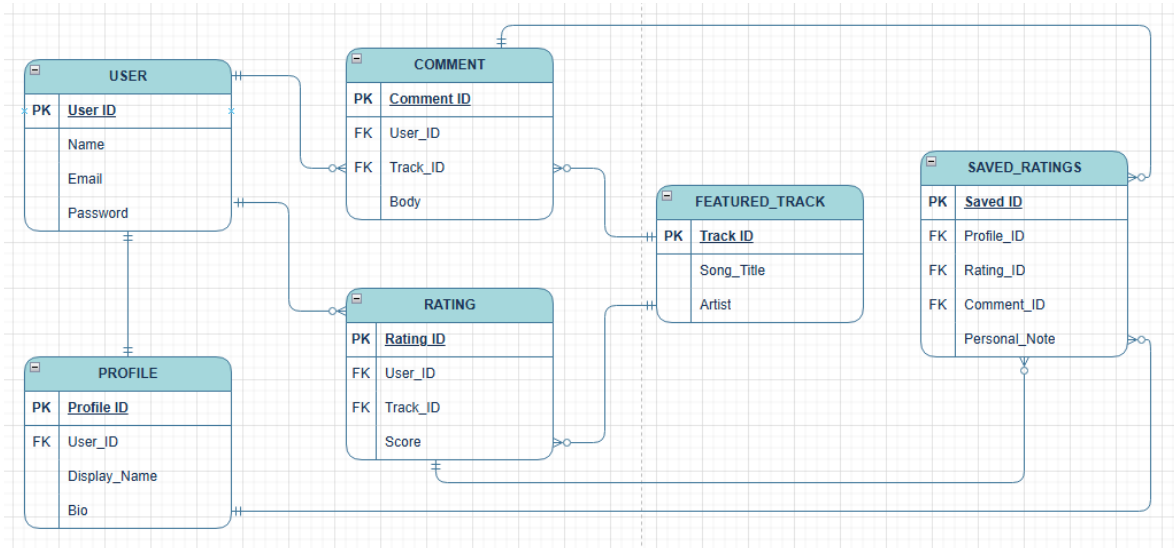
Registered General Users. The primary user category consists of individuals who create accounts to log music, assign ratings, and compose reviews. These users interact with the platform's full feature set and represent the core community of SoundLog. They are assumed to possess basic digital literacy and familiarity with social media conventions.

Guest or Unregistered Visitors. A secondary user category encompasses individuals who access SoundLog without creating an account. Guest users are afforded read-only access to public profiles, reviews, and music entries, facilitating discovery and encouraging registration. No logging, rating, or social features are available to this category.

Content Moderators and Administrators. System administrators are responsible for managing the integrity of the music database, moderating user-generated content in accordance with community guidelines, and performing system maintenance. Administrator accounts possess elevated permissions not available to general users.

Power Users and List Curators. A subset of registered users who engage deeply with the platform's list-creation and curation features, assembling thematic collections of albums or tracks. These users may attract followers due to the quality of their curated content and reviews.

2.3 ERD



2.4 Tools Used

The analysis and design of SoundLog draws upon a combination of industry-standard methodologies, software engineering tools, and design frameworks. The following tools and technologies were employed or specified in the course of this project:

Design and Prototyping Tools

Figma. Utilized for the creation of wireframes, high-fidelity mockups, and interactive prototypes of the SoundLog user interface. Figma's collaborative environment supports iterative design review and stakeholder feedback.

draw.io (diagrams.net). Employed for the construction of system diagrams, including use case diagrams, entity-relationship diagrams (ERDs), and data flow diagrams (DFDs) in accordance with standard UML notation.

Front-End Technologies

React.js. A JavaScript library designated for building the interactive, component-based user interface of SoundLog. React's virtual DOM architecture facilitates efficient rendering of dynamic content, including user feeds, review lists, and music entries.

Tailwind CSS. A utility-first CSS framework adopted for rapid, consistent, and responsive styling of the SoundLog interface, enabling adherence to a cohesive visual identity across components.

Back-End Technologies

Node.js with Express.js. The server-side runtime and framework designated for handling API requests, business logic execution, and middleware processing within the SoundLog architecture.

PostgreSQL. A robust relational database management system selected for the persistent storage of user data, music entries, reviews, ratings, and social relationships due to its ACID compliance and support for complex queries.

JSON Web Tokens (JWT). Employed for stateless, secure user authentication, enabling the server to verify user identity across API requests without maintaining session state.

Development and Version Control

Git and GitHub. Version control and collaborative development are managed through Git, with the remote repository hosted on GitHub to facilitate change tracking, branching strategies, and code review processes.

Visual Studio Code. The primary integrated development environment (IDE) utilized for code authoring, debugging, and extension-supported development workflows.

2.5 Functional Requirements

Functional requirements define the specific behaviors, operations, and services that SoundLog must provide to its users. The following requirements have been derived from the objectives stated in Chapter 1 and from analysis of comparable cataloging platforms:

User Registration and Authentication	The system shall provide a user registration interface requiring a valid email address, a unique username, and a secure password. The system shall authenticate registered users through a secure login mechanism utilizing JWT-based session management. Users shall be able to reset their passwords via an email verification process.
Music Entry Cataloging	The system shall maintain a structured database of music entries encompassing albums, individual tracks, and artists. Each entry shall store metadata including title, artist name, genre, release year, album artwork, and track listing where applicable. Administrators shall be capable of adding, editing, and removing entries

	from the catalog.
Listening Log Management	Registered users shall be able to log music entries to their personal listening history. The log shall record the date of logging, whether the entry has been listened to, and any associated notes. Users shall be able to edit or remove entries from their personal log at any time.
Rating and Review System	The system shall permit registered users to assign a numerical star rating (on a scale of 0.5 to 5.0 in increments of 0.5) to any music entry in the catalog. Users shall be able to compose, edit, and delete written reviews associated with their ratings. Reviews shall be publicly visible on the relevant music entry page and the user's profile.
Social Interaction Features	The system shall permit registered users to assign a numerical star rating (on a scale of 0.5 to 5.0 in increments of 0.5) to any music entry in the catalog. Users shall be able to compose, edit, and delete written reviews associated with their ratings. Reviews shall be publicly visible on the relevant music entry page and the user's profile.
Curated List Creation	Registered users shall be able to create named, thematic lists of music entries. Lists shall be assignable with a title, description, and privacy setting (public or private). Public lists shall be viewable by all users and discoverable through search.
Search and Discovery	The system shall provide a full-text search capability enabling users to locate music entries by title, artist, or genre, as well as to search for user profiles by

	username. Advanced filtering options shall allow users to narrow search results by genre, release year range, and average community rating.
User Profile Management	Each registered user shall have a publicly viewable profile displaying their listening statistics, recent logs, published reviews, and curated lists. Users shall be able to edit their profile information, including display name, biography, and profile picture. Privacy settings shall allow users to set their profile visibility to public or private.

2.6 Non-functional Requirements

Non-functional requirements specify the quality attributes and constraints governing the overall behavior, performance, and integrity of the SoundLog system. These requirements are critical to ensuring user satisfaction and system reliability beyond the scope of individual features.

Performance	The system shall return search query results within two seconds under normal load conditions (up to 500 concurrent users). Page load times for standard content pages shall not exceed three seconds on a stable broadband connection. Database queries shall be optimized through appropriate indexing to maintain consistent response times as the volume of music entries and user records scales.
Security	All user passwords shall be hashed using a recognized cryptographic algorithm (bcrypt) prior to storage. HTTPS shall be

	enforced across all communication between the client and server to prevent data interception. The system shall implement input validation and parameterized queries to protect against SQL injection and cross-site scripting (XSS) attacks. JWT tokens shall have configurable expiry durations to minimize the risk of unauthorized access.
Usability	The SoundLog interface shall conform to established principles of user interface design, ensuring intuitive navigation, consistent visual language, and accessible interaction patterns. The platform shall comply with WCAG 2.1 Level AA accessibility guidelines to accommodate users with visual and motor impairments. Onboarding flows shall require no more than three steps to complete initial account setup and first music log entry.
Scalability	The system architecture shall be designed to support horizontal scaling of server instances to accommodate growth in the user base and concurrent usage. The database schema shall be structured to efficiently manage a catalog of up to one million music entries without requiring fundamental architectural revision. The codebase shall adhere to modular design principles to facilitate the incremental addition of features without systemic disruption.
Reliability and Availability	SoundLog shall target a system uptime of 99.5% or higher under standard operational conditions. Automated database backups shall be performed on a daily schedule to prevent data loss. Error logging and monitoring services shall be integrated to facilitate rapid identification and resolution of system faults.
Maintainability	The system codebase shall follow documented coding standards, including consistent naming conventions, modular component architecture, and inline documentation. All API endpoints shall be

	documented using the OpenAPI specification to facilitate future development and third-party integration. The system shall be designed to permit the deployment of updates without requiring full system downtime.
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APPENDICES

